

Modern Algebra 1

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Lecture 1: Syllabus Day

Tue 03 Sep 2024 09:30

Chapter 1

Introduction to Groups

1.1 Applications

Applied

- Physics & chemistry
- Comp sci - cryptography (Particularly RSA, ECC)
- Robotics??? Modelinng movements
- Economics??? Symmetries in games, game theory

Pure

- Symmetries of roots of polynomials, Galois
- Representation theory, relates groups to lin alg
- Symmetries in geometry & topology

1.2 Symmetries

Definition 1.1: Symmetry

A symmetry of a geometric object is a rearrangement of the figure preserving all properties (the arrangements of sides, vertices, distances, and angles).

For example, a 60/60/60 triangle can be rotated by 120 degrees without changing the shape, or it can be flipped directly about one of its vertices. Both preserve all geometric properties. These transformations are rotation and reflection, respectively. Translation technically works but they don't count cuz boring lol. However doing nothing to the triangle (identity transformation), it's symmetric about that transformation. Oh and flipping about a line (equiv to 180 deg rotation) isn't symmetric. However, we can then rotate it another 180 degrees to obtain a symmetry.

Claim. The only symmetries of a triangle are the identity, 2 rotations and 3 reflections.

Proof. Each symmetry is determined by the different possible locations of each *specific* vertex, and they can have 2 orientations (face up or down), and 3 locations per orientation. $3 \cdot 2 = 6$. ■

— **Remark** —

This group of symmetries, as we will learn later, is the dihedral group D_3 .

We can compose symmetric transformations, giving rise to another symmetry. wow its almost as if its a group...

Call the rotation by 60 degrees transformation R , and call the reflection transformation S . Then we can compose functions:

SR is a symmetry..

RR is a symmetry.

SS is a symmetry..

etc

Definition 1.2: Cayley Table

The Cayley Table of a group (of symmetries) is a table indexed by symmetries as rows and columns, whose entries in the row A and column B is the symmetry BA .

Cayley Table for D_3						
R	R	S	RR	I	RS	RRS
S	1	2	3	4	5	6
RR	1	2	3	4	5	6
I	1	2	3	4	5	6
RS	1	2	3	4	5	6
RRS	1	2	3	4	5	6

To standardize the definition of a rotation and reflection, let's look at the symmetries of a square. We should find 8 symmetries (2 orientations, 4 vertices, $4 * 2 = 8$).

- Rotate 90 degrees R_{90}
- Rotate 180 degrees R_{180}
- Rotate 270 degrees R_{270}
- Rotate 0 degrees 1
- Reflect and rotate 0 degrees S
- Reflect and rotate 90 degrees $R_{90}S$
- Reflect and rotate 180 degrees $R_{180}S$
- Reflect and rotate 270 degrees $R_{270}S$

Cayley Table for the group D_4 , represented with unconventional notation.

	R_0	R_{90}	R_{180}	R_{270}	H	V	D	D'
R_0	R_0	R_{90}	R_{180}	R_{270}	H	V	D	D'
R_{90}	R_{90}	R_{180}	R_{270}	R_0	D'	D	H	V
R_{180}	R_{180}	R_{270}	R_0	R_{90}	V	H	D'	D
R_{270}	R_{270}	R_0	R_{90}	R_{180}	idk	idk	idk	idk
H	H	D	V	D'	R_0	R_{180}	R_{90}	R_{270}
V	V	D'	H	D	R_{180}	R_0	R_{270}	R_{90}
D	D	V	D'	H	R_{270}	R_{90}	R_0	R_{180}
D'	D'	H	D	V	R_{90}	R_{270}	R_{180}	R_0

This table has a few specific properties:

- This table is filled in without introducing new properties (closure).
- Each symmetry can be represented as a composition of a standard 90 degree rotation r and a standard reflection s (basis of dihedral group).
- Everything times R_0 stays the same; $AR_0 = R_0A = A$ (identity element).

— **Remark** —

The elements do not necessarily commute.

Lecture 2: Review of Proofs

Thu 05 Sep 2024 09:30

No discussion tomorrow - and they will all be canceled until the grad students get tf off strike

Notation. For $a, b \in \mathbb{Z}$, if a divides b , that is $b/a \in \mathbb{Z}$, then we write $a|b$ to mean a divides b .

1.3 Integers Mod n

Definition 1.3: Integer Equivalence mod n

Integers $a, b \in \mathbb{Z}$ are equivalent mod n if n divides $a - b$ (the remainders are the same), and we write

$$a \equiv b \pmod{n}.$$

For example, $7 + 4 \equiv 1 \pmod{5}$, since $5|(11 - 1)$.

Definition 1.4: Integers modulo n

The set of integers modulo n is the set $\{0, \dots, n - 1\}$, and is denoted $\mathbb{Z}_n$. In $\mathbb{Z}_n$, addition and multiplication are done modulo n .

The Cayley Table for $\mathbb{Z}_6$

$+$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

Proposition 1.1

Let $\mathbb{Z}_n$ be the set of integers modulo n , and let $a, b, c \in \mathbb{Z}_n$. We have

- $a + (b + c) = (a + b) + c \pmod{n}$.
- There exists an additive identity 0 such that for all $a \in \mathbb{Z}_n$, $a + 0 = a \pmod{n}$.
- For every $a \in \mathbb{Z}_n$, there exists an additive inverse $-a \in \mathbb{Z}_n$ such that $a + (-a) = -a + a = 0 \pmod{n}$.
- $a + b = b + a \pmod{n}$ for all $a, b \in \mathbb{Z}$.

Proof. (1) Since $(a + b) + c = a + (b + c)$ in the integers, then the remainders mod n are also equal. The rest of the proof is left as an exercise ■

Chapter 2

Groups

Definition 2.1: Binary Operation

Let G be a set. A binary operation on G is a function that assigns each ordered pair of elements of G to an element of G .

$$\cdot : G \times G \rightarrow G.$$

For example, in the case of $G = D_4$, then function composition $\circ(A, B) = BA$ is a binary operation on G . If $G = \mathbb{Z}_n$, then the binary operation is addition $+(a, b) = a + b \pmod{n}$.

Definition 2.2: Group

Let G be a set together with a binary operation under which G is closed:

$$\cdot : G \times G \rightarrow G$$

$$\cdot : (a, b) \mapsto ab.$$

We say that G is a group under this operation if the following properties are satisfied:

1. Associativity - for any $a, b, c \in G$, $a(bc) = (ab)c$.
2. Identity - there exists some $e \in G$ such that for all $g \in G$, $ge = eg = g$.
3. Inverses - for all $a \in G$, there exists a corresponding $b \in G$ such that $ab = ba = e$. This is usually denoted a^{-1} .

Definition 2.3: Abelian Group

Let G be a group. We call G an **abelian** group if $ab = ba$ for all $a, b \in G$ (commutative property). Otherwise, the group is non-abelian.

For example, D_4 under function composition is called the Dihedral group of order 8, and $\mathbb{Z}_n$ under addition mod n is the group of integers mod n . D_4 is non-abelian, while $\mathbb{Z}_n$ is abelian. More examples:

- $\mathbb{Z}$ under addition is a group.
- $\mathbb{Z}$ under division is **not** a group.
- $\mathbb{Z}$ under multiplication is **not** a group.
- $\mathbb{R}^*$ (the set of nonzero reals) is a group under multiplication.
- $M_2(\mathbb{R})$ (set of 2×2 matrices with real entries) is a group under addition.

- $\text{GL}_2(\mathbb{R}) \subseteq M_2(\mathbb{R})$ the general linear is a group under multiplication.

QUATERNIONS!

Let 1 be the identity matrix,

$$I = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \quad K = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix},$$

where $i^2 = -1$. Then

$$I^2 = J^2 = K^2 = -1, \quad IJ = K, \quad JK = -I, \quad KI = J, \quad JI = -K,$$

$$KJ = -I, \quad IK = -J.$$

The group $\{\pm 1, \pm I, \pm J, \pm K\}$ is known as the quaternion group under multiplication.

Lecture 3: Isomorphisms, Day 2

Mon 07 Oct 2024 20:37

— As Previously Seen —

Two groups are isomorphic $G \cong H$ if there exists a bijection $\phi : G \rightarrow H$ such that $\phi(ab) = \phi(a)\phi(b)$ for all $a, b \in G$. Every infinite cyclic group is isomorphic to $\mathbb{Z}$ and every finite cyclic group of order n is isomorphic to $\mathbb{Z}_n$. Cayley's theorem states every group is isomorphic to the group of permutations of some set.

Theorem 2.1 (Cayley)

Every group is isomorphic to a group of permutations of some set.

Proof. Let G be a group. We want to find a group H of permutations such that $G \cong H$. For all $g \in G$, define $Tg : G \rightarrow G$ as the map $a \mapsto ga$. We claim Tg is a permutation of G , and

$$H = \{Tg \mid g \in G\}.$$

First, we show Tg is a permutation. We must show Tg is a bijection on G . Let $a \in G$. We have $a = gg^{-1}a = Tg(g^{-1}a)$. Therefore, Tg is surjective. Now let $Tg(a) = Tg(b)$ for some $a, b \in G$. It follows that

$$ga = gb \implies a = b.$$

Therefore, Tg is bijective, and is thus a permutation of G .

Next, we show H is a group. Let $a, b, g \in G$. We have

$$(TaTb)(g) = Ta(bg) = abg = (ab)g = Tab(g).$$

Since $ab \in G$, $Tab \in H$. Let $a, b, c, g \in G$. We have

$$((TaTb)Tc)(g) = (abc)g.$$

Associativity then follows from the fact that G is a group. Let $e \in G$ be the identity. Let $a, g \in G$. It follows that

$$(TaTe)(g) = aeg = ag = Ta(g).$$

We say Te is the identity. Finally, let $a, a^{-1} \in G$ be inverses. We have

$$(TaTa^{-1})(g) = aa^{-1}g = eg = Te(g).$$

Therefore, inverses exist, and H is a group.

Next, we show $G \cong H$. Let $\phi : G \rightarrow H$ be the map $g \mapsto Tg$. We need only show ϕ is an isomorphism. By definition of H , ϕ is surjective. We need only show ϕ is injective and homomorphic. Suppose $\phi(a) = \phi(b)$. We then have

$$Ta(g) = Tb(g) \implies ag = bg \implies a = b.$$

Therefore, ϕ is injective. Finally, we show homomorphism.

$$\phi(ab)(g) = (TaTb)(g) = (ab)g = Tab(g) = \phi(ab)(g).$$

Therefore, ϕ is a homomorphism, and thus an isomorphism. ■

2.0.1 Equivalence Relations (Ch. 0)

Equivalence relations generalize the idea of equality. For example, although 5 and 12 are different numbers, $5 \equiv 7 \pmod{7}$. So we have some sense of equality, but they're not equal. Equivalence relations generalize equality to give us a way of saying this.

Definition 2.4: Equivalence Relation

An equivalence relation on a set S is a set R of ordered pairs of elements S , $R \subseteq S \times S$, such that

- (Reflexive) $(a, a) \in R$ for all $a \in S$.
- (Symmetric) If $(a, b) \in R$, then $(b, a) \in R$.
- (Transitive) If $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$.

If $(a, b) \in R$, we write $a \sim b$, $a \approx b$, or $a \equiv b$. We usually just use $a \sim b$.

Definition 2.5: Equivalence Classes

For some $a \in S$, define

$$[a] := \{b \in S \mid (a, b) \in R\}.$$

We call $[a]$ the equivalence class of a , and any $b \in [a]$ has $b \sim_R a$.

For example, $\mathbb{Z}_n = \{[a] \mid a \in \mathbb{Z} \wedge 0 \leq a < n\}$ for $R = \{(a, b) \in \mathbb{Z}^2 \mid a \equiv b \pmod{n}\}$.

Here's a great example. Let

$$S := \{(a, b) \mid a, b \in \mathbb{Z} \wedge b \neq 0\}.$$

Then the rational numbers are defined via the equivalence relation

$$R = \{((a, b), (c, d)) \in S^2 \mid b, d \neq 0 \wedge ad = bc\}.$$

Saying $((a, b), (c, d)) \in R$ is equivalent to saying

$$\frac{a}{b} = \frac{c}{d}.$$

I will prove this later for fun i guess

Definition 2.6: Partition

A **partition** of a set S is a collection $P = \{S_1, \dots, S_{|S|}\}$ of nonempty disjoint subsets of S such that

$$\bigcup_{i=1}^{|S|} S_i = S.$$

For example, the sets of even and odd integers partition $\mathbb{Z}$.

Theorem 2.2

Let R be an equivalence relation $\sim$ on a set S . Then the equivalence classes from R form a partition of S . Conversely, for any partition P on S , there is an equivalence relation on S where the equivalence classes are the elements of P .

Chapter 3

Cosets and Lagrange's Theorem

Definition 3.1: Coset

Let G be a group, and let $H \subseteq G$ be a nonempty subset of G . For any $a \in G$, we define the sets

$$aH = \{ah \mid h \in H\},$$

and

$$Ha = \{ha \mid h \in H\}.$$

When H is a subgroup of G , we call aH the *left coset* of H in G containing a , and similarly, Ha is the *right coset* of H in G containing a . We call a the coset representative.

Notation. $|aH|$ is the number of elements in aH . Same for Ha . This follows since aH is not generally a subgroup, so we instead have the order or cardinality of aH . Additionally, aHa^{-1} is the set

$$aHa^{-1} = \{aha^{-1} \mid h \in H\}.$$

For example, let $G = S_3$, and let $H = \{e, (23)\}$. Note that $e \in G$. We then have

$$eH = H.$$

Note also that $(12) \in G$. We then have

$$(12)H = \{(12)e, (12)(23)\} = \{(12), (123)\}.$$

Finally, note $(13) \in G$. We have

$$(13)H = \{(13)e, (13)(23)\} = \{(13), (132)\}.$$

Interestingly, we can check by hand that these are the only left cosets of G .

Proposition 3.1 (Properties of Cosets)

Let $H \leq G$, $a, b \in G$. Then the following are true.

- $a \in aH$.
- $aH = H$ iff $a \in H$.
- $(ab)H = a(bH)$ and $H(ab) = (Ha)b$.
- $aH = bH$ if and only if $a \in bH$.
- Either $aH = bH$ or $aH \cap bH = \emptyset$.
- $aH = bH$ if and only if $a^{-1}b \in H$.
- $|aH| = |bH|$ for all $a, b \in G$.
- $aH = Ha$ if and only if $H = aHa^{-1}$.
- aH is a subgroup of G if and only if $aH = H$.

Proof. 1,2,3 easy. 4 $aH = bH \implies ae = bh$ for some $h \in H$ since $e \in H$. then $a \in bH$. Then $a = bh$ for some $h \in H$, so $aH = bhH = bH$ for $h \in H$. the rest i do on my own. ■

Lecture 4: Lagrange's Theorem

Thu 10 Oct 2024 09:32

Definition 3.2: Index

Let G be a group with $H \leq G$. Then the index of H in G , denoted $[G : H]$, is denoted

$$[G : H] = \frac{|G|}{|H|}.$$

Lagrange's theorem is an extremely important result in group theory, and justifies our study of these seemingly arbitrary "cosets". One of the immediate corollaries of Lagrange's theorem generalizes one extremely useful property of cyclic groups, being that the order of their subgroups divides the order of the group. Many other important theorems also follow from the result, and Lagrange's theorem will remain important throughout our study of algebra.

Theorem 3.1 (Lagrange)

Let G be a finite group with subgroup $H \leq G$. Then $[G : H]$ is the number of distinct left cosets of H in G . In particular, the order of H divides the order of G .

Proof. Let $\{g_1, \dots, g_k\} \subseteq G$ such that $g_1H, \dots, g_kH$ are *all of the distinct* cosets of H in G . We will show $\{g_1H, \dots, g_kH\}$ is a partition on G . By Proposition idk, for any g_iH, g_jH for $i \neq j$, $g_iH \cap g_jH = \emptyset$. Now let $a \in G$, and suppose there does not exist some g_iH such that $a \in g_iH$. Since H is a subgroup, $e \in H$, and thus $ae = a \in aH$. By proposition idk, since $a \notin g_iH$ for all i , we know $aH \neq g_iH$ for all i , and thus the set $\{g_1H, \dots, g_kH\}$ is not the set of *all* distinct cosets of H in G . By contradiction, and by the methods by which we reached a contradiction, we have $a \in g_iH$ for some i , and $\{g_1H, \dots, g_kH\}$

partitions G .

By Proposition idk, note $|g_i H| = |g_j H|$ for all i, j . Since $g_1 H, \dots, g_k H$ forms a partition of G , we have

$$|G| = \sum_{i=1}^k |g_i H| = k |g_i H|.$$

Therefore,

$$\frac{|G|}{|g_i H|} = k,$$

and k is the number of cosets. However, there exists some $g_i = e$, and $eH = H$, so we have

$$\frac{|G|}{|H|} = k \implies [G : H] = k.$$

■

Remark

didn't just define $[G : H]$ the number of cosets and show

$$[G : H] = \frac{|G|}{|H|}?$$

Now we will explore the many corollaries of 3.1.

Corollary 3.1

Let G be a finite group. If $g \in G$, then $|g|$ divides $|G|$.

Proof. Let $H = \langle g \rangle$. Thus, $|H| = |g|$. By 3.1, $|H|$ divides $|G|$, so $|g|$ divides $|G|$. ■

Corollary 3.2

Let G be a finite group. For all $a \in G$, $a^{|G|} = e$.

Proof. By 3.1, for any $a \in G$, $|a| \mid |G|$. Thus,

$$|G| = k |a|$$

for some $k \in \mathbb{Z}$, and thus

$$|a| = \frac{|G|}{k} \in \mathbb{N}.$$

We then have

$$a^{|G|/k} = e.$$

Thus,

$$a^{|G|} = a^{k(|G|/k)} = e^k = e.$$

■

Definition 3.3: Prime Number

A positive number $p \geq 2$ is prime if and only if the positive divisors of p are precisely 1 and p .

Proposition 3.2

Let $p \in \mathbb{N}$. Then p is prime if and only if when $p|ab$, then $p|a$ or $p|b$.

Corollary 3.3

Let $|G| = p$ for a prime number p prime. Then G is cyclic, generated by any $a \in G \setminus \{e\}$.

Proof. Let $a \in G \setminus \{e\}$. Let $H = \langle a \rangle$. It follows that $|H| \geq 2$, since $a \neq e$, so $a, e \in H$. By 3.1, the order of H divides $|G|$. Since $|G|$ is prime, by 3.3, the only divisors of $|G|$ are 1 and $|G|$. Since $|H| > 1$, we have $|H| = |G|$, and thus $H = G$. ■

Corollary 3.4

Let G be a finite group with $|G| = p$ prime. Then $G \cong \mathbb{Z}_p$.

Proof. By 3.3, G is cyclic with order p . By an earlier theorem I still have to prove, $G \cong \mathbb{Z}_{|G|} = \mathbb{Z}_p$. ■

Corollary 3.5

If H and K are subgroups of a finite group G with $K \subset H \subset G$, then

$$[G : K] = [G : H][H : K].$$

Proof.

$$[G : K] = \frac{|G|}{|K|} = \frac{|G|}{|H|} \cdot \frac{|H|}{|K|}.$$

Remark

The converse of Lagrange's Theorem is *not true*. That is, if $b| |G|$, is possible for there to be no subgroups of order b .

Theorem 3.2 (Fermat's Little Theorem)

Let p be a prime number and suppose p does not divide a . Then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Equivalently,

$$a^p \equiv a \pmod{p}.$$

Proof. Since a does not divide p , we know $a = pm + r$ for $0 < r < p$. We then know

$$a^{p-1} \equiv (pm + r)^p \equiv r^{p-1} \pmod{p}$$

by like 10 different theorems, such as the binomial theorem. We can then assume $r \in U(p)$, since $r \neq p$. Since $|U(p)| = p - 1$ then $r^{p-1} \equiv 1 \pmod{p}$. ■

Theorem 3.3 (Generalized Fermat)

Let a be relatively prime to n . Then $a^{\phi(n)} \equiv 1 \pmod{n}$.

Proof. Since a is relatively prime to n , then $a \in U(n)$. Note that $\phi(n) = |U(n)|$. Thus, $a^{\phi(n)} = a^{|U(n)|}$. By 3.1, $|a|$ divides $|U(n)|$, and thus $a^{|U(n)|} \equiv 1 \pmod{n}$. ■

3.1 Application to Permutation Groups

This is a window into “group actions”. The book does not cover group actions, so I’ll ask the prof for resources about it.

Definition 3.4: Stabilizer and Orbit

Let G be a group of permutations of some set S . Then for each $i \in S$, define the *stabilizer* of i in G ,

$$\text{stab}_G(i) = \{\phi \in G \mid \phi(i) = i\},$$

and the *orbit* of i in G ,

$$\text{orb}_G(i) = \{\phi(i) \mid \phi \in G\}.$$

Theorem 3.4

$\text{stab}_G(i) \leq G$. $\text{orb}_G(i) \subseteq S$.

Proof. Let G be a group of permutations of S , and let $i \in S$. First, we show $\text{stab}_G(i)$ is closed under function composition. Let $\phi, \psi \in \text{stab}_G(i)$. Since all $\phi \in \text{stab}_G(i)$, have $\phi(i) = i$, it follows that

$$(\phi\psi)(i) = \phi(\psi(i)) = \phi(i) = i.$$

Therefore, $\phi\psi \in \text{stab}_G(i)$.

Associativity follows from the fact that $\text{stab}_G(i) \subseteq G$, which is a group. Notice that the identity map $\mathbb{1}_S$ maps $i \mapsto i$, so $\mathbb{1}_S \in \text{stab}_G(i)$. Finally, we show inverses exist. Let $\phi \in \text{stab}_G(i)$, and let ϕ^{-1} be its inverse in G . It follows that

$$(\phi^{-1}\phi)(i) = i.$$

However, $\phi(i) = i$, so it follows that

$$\phi^{-1}(\phi(i)) = \phi^{-1}(i) = i.$$

Therefore, ϕ^{-1} fixes i , and thus $\phi^{-1} \in \text{stab}_G(i)$. Thus, $\text{stab}_G(i)$ is a group. Since $\text{stab}_G(i) \subseteq G$, we have $\text{stab}_G(i) \leq G$.

The second fact is trivial. For any $\phi \in G$, we know ϕ is by definition a bijection $S \hookrightarrow S$, so $\phi(i) \in S$ for all $i \in S$. Therefore, any $i' \in \text{orb}_G(i)$ has $i' = \phi(i)$ for some $\phi \in G$, and thus $i' \in S$. Therefore, $\text{orb}_G(i) \subseteq S$. ■

Theorem 3.5 (Orbit-Stabilizer)

Let G be a finite group of permutations of a set S . Then for any $i \in S$,

$$|G| = |\text{orb}_G(i)| |\text{stab}_G(i)|.$$

Proof. this one is harder ■

Chapter 4

External Direct Products

Lecture 5: Direct Products

Thu 17 Oct 2024 09:30

Direct products allow us to construct new groups from previous groups, particularly through the cartesian product of these groups.

Definition 4.1: Cartesian Product

The **cartesian product** of two sets S and R is the set of all (s, r) such that $s \in S$ and $r \in R$.

$$S \times R = \{(s, r) \mid s \in S \wedge r \in R\}.$$

The cartesian product of a set with itself is typically denoted $R \times R = R^2$. Now we consider the application of the cartesian product to groups. Let $(G, \cdot)$ and $(H, \circ)$ be groups. We define the binary operation

$$\star : (G \times H) \times (G \times H) \rightarrow G \times H$$

as the map

$$\star : ((g_1, h_1), (g_2, h_2)) \mapsto (g_1 \cdot g_2, h_1 \circ h_2).$$

We will use juxtaposition instead of writing out the operators from here out.

Proposition 4.1

Let G, H be groups. Let $G \times H$ have associated with it the binary operation

$$(g_1, h_1) \cdot (g_2, h_2) = (g_1 g_2, h_1 h_2).$$

Then $G \times H$ is a group.

Proof. The binary operation is closed since G and H are groups. The identity is (e_G, e_H) for e_G the identity in G and e_H the identity in H . The inverse of some $(g, h) \in G \times H$ is $(g^{-1}, h^{-1}) \in G \times H$. Associativity follows from associativity. ■

Definition 4.2: External Direct Product

The group $G \times H$ is called the external direct product of G and H , but is really just called the **direct product** of G and H .

Notation. The book writes $G \oplus H$ for direct products. This is technically the “direct sum”, which coincides with $\times$ for finitely many groups. However, people generally write $G \times H$ for direct products, not $G \oplus H$.

If we continue the same construction, we can define a direct product for n groups. Let $G_1, \dots, G_n$ be groups. Then

$$G = \prod_{i=1}^n G_i$$

is the direct product of these groups.

Definition 4.3: Least Common Multiple

The least common multiple of two numbers $a, b \in \mathbb{Z} \setminus \{0\}$ is the smallest positive integer that is a multiple of a and b .

$$\text{lcm}(a, b) = \max \{n \in \mathbb{Z}^+ \mid \exists k \in \mathbb{Z} : n = ka \wedge \exists \ell \in \mathbb{Z} : n = \ell b\}.$$

Proposition 4.2

The least common multiple of $a, b \in \mathbb{Z} \setminus \{0\}$ is

$$\text{lcm}(a, b) = \frac{ab}{\gcd(a, b)}.$$

Theorem 4.1

Let $(g, h) \in G \times H$. Then

$$|(g, h)| = \text{lcm}(|g|, |h|).$$

In general, let

$$(g_1, \dots, g_n) \in \prod_{i=1}^n G_i.$$

Then

$$|(g_1, \dots, g_n)| = \text{lcm}(|g_1|, \dots, |g_n|).$$

Proof. EFY ■

Theorem 4.2

$$\mathbb{Z}_n \times \mathbb{Z}_m \cong \mathbb{Z}_{nm} \iff \gcd(n, m) = 1.$$

Proof. We first show $\mathbb{Z}_n \times \mathbb{Z}_m \cong \mathbb{Z}_{nm}$ implies that $\gcd(n, m) = 1$. We will prove by contrapositive. Suppose $\gcd(n, m) \neq 1$. We will show there is no isomorphism.

Let $d = \gcd(n, m) > 1$. Then $\frac{mn}{d}$ is divisible by both m and n , hence for any element $(a, b) \in \mathbb{Z}_m \times \mathbb{Z}_n$,

$$\frac{mn}{d}(a, b) = (0, 0).$$

Since $\frac{mn}{d} < mn$, then $\mathbb{Z}_m \times \mathbb{Z}_n$ is not cyclic.

Now we show if $\gcd(n, m) = 1$, then an isomorphism exists. Let $\gcd(n, m) = 1$. Since there exists

$1 \in \mathbb{Z}_m$ with $|1| = m$, and likewise for n , there exists $(1, 1) \in \mathbb{Z}_m \times \mathbb{Z}_n$ such that, by thm:4.1,

$$|(1, 1)| \frac{mn}{\gcd(n, m)} = mn.$$

Therefore, $\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn}$. ■

Corollary 4.1

Let $n_1, \dots, n_k \in \mathbb{Z}^+$. Then

$$\prod_{i=1}^k \mathbb{Z}_{n_i} \cong \mathbb{Z}_{\prod_{i=1}^k n_i}$$

if and only if $\gcd(n_i, n_j) = 1$ for all i, j .

Proof. EFY ■

Corollary 4.2

Let $m = p_1^{e_1} \cdots p_k^{e_k}$ for p_i primes. Then

$$\mathbb{Z}_m \cong \prod_{i=1}^k \mathbb{Z}_{p_i^{e_i}}.$$

Proof. EFY ■

Chapter 5

Normal Subgroups and Factor Groups

We begin with discussion of internal direct products.

Definition 5.1: Internal Products

Let G be a group with subgroups H and K satisfying the following conditions:

- $G = HK = \{hk \mid h \in H \wedge k \in K\}$.
- $H \cap K = \{e\}$.
- $hk = kh$ for all $k \in K$ and $h \in H$.

Then G is the internal product of H and K .

Note. Not every group can be written as the internal product of two proper subgroups. For example, S_3 cannot.

Theorem 5.1

Let G be the internal product of $H, K \leq G$. Then $G \cong H \times K$.

Lecture 6: Fundamental Theorem of Finite Abelian Groups

Tue 22 Oct 2024 09:30

Proof. Let $\phi : G \rightarrow H \times K$. Since $G = HK$, we know for all $g \in G$, there exists $h \in H$ and $k \in K$ such that $g = hk$. As such, let $g \mapsto (h, k)$. It's not apparent that g has a unique representation as hk , so we must show this is true. Suppose $g = h_1k_1 = h_2k_2$. We must show $h_1 = h_2$ and $k_1 = k_2$. Since $h_1k_1 = h_2k_2$, it follows that

$$h_2^{-1}h_1 = k_2k_1^{-1}.$$

Since $h_2^{-1}h_1 \in H$ and $k_2k_1^{-1} \in K$, we have $h_2^{-1}h_1 \in H \cap K = \{e\}$. By the same logic, we have $h_1 = h_2$ and $k_1 = k_2$.

Next, we show ϕ is a homomorphism. Let $h_1k_1, h_2k_2 \in G$. Then

$$\begin{aligned}\phi(h_1k_1)\phi(h_2k_2) &= (h_1, k_1)(h_2, k_2) \\ &= (h_1h_2, k_1k_2) \\ &= \phi(h_1h_2k_1k_2) \\ &= \phi(h_1k_1h_2k_2).\end{aligned}$$

Therefore, ϕ is a homomorphism. Now we show ϕ is injective. Let $\phi(g_1) = \phi(g_2)$, and notice it follows that $(h_1, k_1) = (h_2, k_2)$. Therefore, $h_1 = h_2$ and $k_1 = k_2$. Thus, $h_1k_1 = h_2k_2$, and since the map is well defined, $g_1 = g_2$. Thus, ϕ is injective.

Finally, we show ϕ is surjective. Let $(h, k) \in H \times K$. It follows that $\phi(hk) = (h, k)$. Since $h \in H$ and $k \in K$, $hk \in HK = G$. Thus, there exists $hk \in G$ such that $\phi(hk) = (h, k)$, and ϕ is surjective. Therefore, $G \cong H \times K$. ■

Theorem 5.2 (Fundamental Theorem of Finite Abelian Groups)

Every finite abelian group G is isomorphic to a direct product of cyclic groups of prime power order. That is,

$$G \cong \prod_{i=1}^n \mathbb{Z}_{p_i^{e_i}},$$

for p prime. Moreover, the number of terms in the product and the orders of the cyclic groups are uniquely determined up to isomorphism.

Proof. h ■

Corollary 5.1

Let p be a prime number. Then $\mathbb{Z}_p \times \mathbb{Z}_p \not\cong \mathbb{Z}_{p^2}$. Additionally, for any $A \in \mathbf{Ab}$, either $A \cong \mathbb{Z}_p \times \mathbb{Z}_p$, or $A \cong \mathbb{Z}_{p^2}$.

Proof. Trivial, EFY. ■

Definition 5.2: Klein Group

The Klein group is the group $\mathbb{Z}_2 \times \mathbb{Z}_2$, and it is the smallest group that is not cyclic up to isomorphism.

Lecture 7: Normal Subgroups and Factor/Quotient Groups

Thu 24 Oct 2024 09:38

Definition 5.3: Normal Subgroup

A subgroup $H \leq G$ is *normal* if $gH = Hg$ for all $g \in G$. This is denoted $H \triangleleft G$.

Proposition 5.1

If G is an abelian group, then all subgroups $H \leq G$ are normal.

Proof. Since $gh = hg$ for all $g, h \in G$, we have $gH = Hg$. ■

Theorem 5.3

Let G be a group with subgroup $N \leq G$. Then the following statements are equivalent:

1. N is a normal subgroup of G .
2. For all $g \in G$, $gNg^{-1} \subseteq N$.
3. For all $g \in G$, $gNg^{-1} = N$.

Proof. (1 $\implies$ 2) Let $N \triangleleft G$. Let $g \in G$. Thus, for all $n \in N$, there exists $n' \in N$ such that $gn = n'g$. It follows that $gng^{-1} = n' \in N$, and thus $gNg^{-1} \subseteq N$ for all n . Therefore, $gNg^{-1} \subseteq N$.

(2 $\implies$ 3) Let $gNg^{-1} \subseteq N$. For all $n \in N$, there exists $n' \in N$ such that $g^{-1}n(g^{-1})^{-1} = n'$, since the statement holds for all $g \in G$. Thus, $n = gn'g^{-1}$, and $N \subseteq gNg^{-1}$.

(3 $\implies$ 1) Let $gNg^{-1} = N$. It follows that for all $n \in N$, there exists $n' \in N$ such that $gng^{-1} = n'$. It follows that $gn = n'g$ for all $n \in N$. Thus, $N \triangleleft G$. ■

When we worked with cosets, we noted that the cosets not only aren't groups themselves, but they didn't form a group either. One reason we study normal subgroups is because they form a group, called a quotient group, or a factor group according to the book because why not.

Consider some $H \leq G$, and consider the set of left cosets $\mathcal{L} = \{gH \mid g \in G\}$. We know $|\mathcal{L}| = [G : H]$. We can define a binary operation on this set:

$$(g_iH)(g_jH) \mapsto g_i g_j H.$$

If this forms a group, that would give us a lot more information about the cosets in the group. Note that eH is the identity, and for any $gH \in \mathcal{L}$, there exists an inverse $g^{-1}H \in \mathcal{L}$. Associativity and closure follow from G being a group. BUT. This map is not well defined. If there exist $g_1, g_2 \in H$ such that $g_1 g_2 H$ does NOT have a unique coset representative in G , then the map is not well defined. We will show this more rigorously.

Define $L \times L \rightarrow L$ as the map $((g_1H), (g_2H)) \mapsto g_1 g_2 H$. Suppose $aH = bH$, and $cH = dH$. If this map was well defined, we would need $aHcH = bHdH$. It would follow that $acH = bdH$. Since $aH = bH$, there exists h_1 such that $b = ah_1$. Similarly, $d = ch_2$. It follows that $bdH = ah_1ch_2H$. Since $h_2 \in H$, we have $ah_1ch_2H = ah_1cH$. We must at this point require $ah_1cH = acH$. For this to be true, we would need $h_1cH = cH$ for all $c \in G$. Since this is not true in general, we do not have a well defined map in general.

However, if we let H be normal in G , then it is true, since $h_1c = ch'$ for some h' , since $Hc = cH$, and thus $h_1cH = ch'H = cH$. Therefore, we can define this group, but only if we take the set of cosets for some normal subgroup $N \triangleleft G$.

Definition 5.4: Quotient Group

Let N be a normal subgroup of G . Then the set of left cosets $\mathcal{L} = \{gN \mid g \in G\}$ under the binary operation

$$(aN)(bN) \mapsto abN$$

is called the quotient group of G and N . We denote it as G/N , and has the order $[G : N]$.

Theorem 5.4

The quotient group is a group.

Proof. EFY, read above logic ■

Remark

Note that G/N is the set of equivalence classes given by the equivalence relation $a \sim b$ if $ab^{-1} \in N$.

Note also that $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$.

Chapter 6

Homomorphisms

Lecture 8: Homomorphisms

Tue 29 Oct 2024 09:34

Isomorphisms are fairly restrictive. The requirement that an isomorphism be bijective means not many groups can be isomorphic. However, by dropping the bijection restriction, we can still uncover a large amount of information about relationships between groups.

Definition 6.1: Homomorphism

A map $\phi : G \rightarrow F$ is a group homomorphism if for all $a, b \in G$, $\phi(a)\phi(b) = \phi(ab)$. An isomorphism is then a bijective homomorphism.

For example, consider $G = \langle g \rangle$, and define $\phi : (\mathbb{Z}, +) \rightarrow G$ as the map $n \mapsto g^n$. This map is clearly not bijective for any g with finite order. However, regardless of whether it is bijective or not, the map is a homomorphism.

Another example is the map $\det : \text{GL}_n(\mathbb{R}) \rightarrow \mathbb{R}^*$, which I've previously shown is a group homomorphism. I'm pretty sure this holds for any field in fact. That is, $\det : \text{GL}_n(\mathbb{F}) \rightarrow \mathbb{F}^*$ should be a homomorphism.

Theorem 6.1 (Properties of Homomorphisms)

Let $\phi : G \rightarrow H$ be a group homomorphism, and let $g \in G$.

1. $\phi(e_G) = e_H$.
2. For all $n \in \mathbb{Z}$, $\phi(g^n) = \phi(g)^n$.
3. If $|g| < \infty$, then $|\phi(g)|$ divides $|g|$. If $|G| < \infty$, then $|\phi(g)|$ divides $|\phi(G)|$.
4. If $K \leq G$ is cyclic, then $\phi(K)$ is cyclic in $\phi(G)$. If $K \leq G$ is normal, then $\phi(K)$ is normal in $\phi(G)$. If $K \leq G$ is abelian, then $\phi(K)$ is abelian in $\phi(G)$.
5. If $K \leq H$, then $\phi^{-1}(K) := \{g \in G \mid \phi(g) \in K\} \leq G$. Moreover, if $K \triangleleft H$, then $\phi^{-1}(K) \triangleleft G$.
6. $\phi(Z(G)) \leq Z(\phi(G))$.

Proof. (1) Let e_G be the identity in G . Then, for all $g \in G$, $e_G g = g$. Thus, $\phi(e_G g) = \phi(g) = \phi(e_G)\phi(g)$. Therefore, $\phi(e_G)$ is the identity in H .

(2) Trivial. Induct on it.

(3) The second claim follows from Lagrange's theorem, so it suffices to show the first claim. Take

$n = |g|$ finite. Then

$$\phi(g)^n = \phi(g^n) = \phi(e_G) = e_F.$$

Therefore, the order of $\phi(g)$ divides n .

(4) We prove the case of normal subgroups. $\phi(K) \triangleleft H$ if and only if $a\phi(K)a^{-1} \subseteq \phi(K)$. Let $a \in \phi(G)$. Thus, $a = \phi(g)$ for some $g \in G$. Let $x = \phi(k)$ for some $k \in K$. Then $axa^{-1} = \phi(g)\phi(k)(\phi(g))^{-1} = \phi(ghg^{-1})$. Because $K \triangleleft G$, $ghg^{-1} \in gKg^{-1} = K$. There then exists $k' \in K$ such that $ghg^{-1} = k'$. Therefore, $axa^{-1} = \phi(h') \in \phi(H)$.

(5) EFY

(6) fun

■

Definition 6.2: Kernel

Let $\phi : G \rightarrow F$ be a group homomorphism. Then the kernel of ϕ , denoted $\ker \phi$, is the set

$$\ker \phi := \{g \in G \mid \phi(g) = e_F\} = \phi^{-1}(e_F).$$

Theorem 6.2 (Properties of the Kernel)

Let $\phi : G \rightarrow F$ be a group homomorphism.

1. $\ker \phi \leq G$. Moreover, $\ker \phi \triangleleft G$.
2. $\phi(a) = \phi(b)$ if and only if $a \ker \phi = b \ker \phi$.
3. If $\phi(g) = a$, then $\phi^{-1}(a) = \{g \in G \mid \phi(g) = a\} = g \ker \phi$.
4. If $|\ker \phi| = n < \infty$, then ϕ is injective.

Proof. (1) $\{e_F\} \triangleleft \phi(G)$, so $\phi^{-1}(e_F) = \ker \phi \triangleleft G$ by property (5) of homomorphisms.

(2) Let $\phi(a) = \phi(b)$. Then $\phi\phi(b)^{-1} = e_F$. Since ϕ is a homomorphism, $\phi(ab^{-1}) = e_F$. This happens if and only if $ab^{-1} \in \ker \phi$ by definition. By properties of cosets, $a \ker \phi = b \ker \phi$.

(3) Similar to (2)

(4) EFY

■

Lecture 9: The First Isomorphism Theorem

Thu 31 Oct 2024 09:27

Theorem 6.3 (First Isomorphism Theorem)

Let $\phi : G \rightarrow F$ be a group homomorphism. Then there exists a group isomorphism $\psi : G/\ker \phi \rightarrow \phi(G)$ defined as $g \ker \phi \mapsto \phi(g)$.

Proof. First, we show ψ is well defined. Let $g \ker \phi = g' \ker \phi$. It follows from properties of cosets that $g' \in g \ker \phi$. Therefore, there exists $h \in \ker \phi$ such that $g' = gh$. By definition, $\phi(g') = \phi(gh) = \phi(g)\phi(h)$. Since $h \in \ker \phi$, $\phi(h) = e$. Therefore, $\phi(g') = \phi(g)$. Therefore, ψ is well defined.

Next, we show ψ is injective. Let $\psi(g \ker \phi) = \psi(g' \ker \phi)$. It follows that $\phi(g) = \phi(g')$ by definition of ψ , and thus $\phi(g)(\phi(g'))^{-1} = e$. Since ϕ is a homomorphism, it follows that $\phi(g)(\phi(g')^{-1}) = \phi(g(g')^{-1}) = e$. It follows that $g(g')^{-1} \in \ker \phi$, and by properties of cosets, $g(g')^{-1} \in \ker \phi$ if and only if $g \ker \phi =$

$g' \ker \phi$. Therefore, ψ is injective.

Now we show ψ is surjective. Let $x = \phi(g) \in \phi(G)$. Notice that $g \ker \phi \mapsto \phi(g)$. Therefore, $x = \psi(g \ker \phi)$, and ψ is surjective.

Finally, we show ψ is a homomorphism. Let $g \ker \phi, g' \ker \phi \in G / \ker \phi$. It follows that

$$\psi(g \ker \phi g' \ker \phi) = \psi(gg' \ker \phi) = \phi(gg').$$

Since ϕ is a homomorphism, it follows that

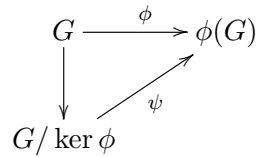
$$\phi(gg') = \phi(g)\phi(g').$$

By definition,

$$\phi(g)\phi(g') = \psi(g \ker \phi)\psi(g' \ker \phi).$$

Therefore, ψ is an isomorphism. ■

The first isomorphism theorem is an extremely powerful theorem, similar to Lagrange's theorem, and deserves some discussion at length.



Note that $\exists! \psi$ such that this diagram commutes.'

Corollary 6.1

Let $\phi : G \rightarrow F$ be a group homomorphism with G a finite group. Then $[G : \ker \phi] = |\phi(G)|$.

Proof.

$$[G : \ker \phi] = \frac{|G|}{|\ker \phi|} = |G / \ker \phi| = |\phi(G)|. \quad \blacksquare$$

Corollary 6.2

Let $\phi : G \rightarrow F$ be a group homomorphism with G finite. Then

$$|\phi(G)| \mid |G| \text{ and } |\phi(G)| \mid |F|.$$

Proof. By the previous corollary, $|G / \ker \phi| = |G|$. Therefore, $|\phi(G)|$ divides $|G|$ trivially. ■

Corollary 6.3

Let $N \triangleleft G$. Then there exists a group homomorphism $\phi : G \rightarrow K$ such that $N = \ker \phi$. Not sure why F follows but its probably trivial.

Proof. Define $\phi : G \rightarrow G/N$ by $g \mapsto gN$. Trivially, this is a homomorphism. Note that the identity is N , and $gN = N$ if and only if $g \in N$ by our properties of cosets. Therefore, $g \mapsto N$ if and only if $g \in N$,

and $\ker \phi = N$. ■

Remark

Normal subgroups are kernels of group homomorphisms, and the kernel of a group homomorphism is a normal subgroup.

Corollary 6.4

The first isomorphism is not required, but it makes it simple. Let $\phi : G \rightarrow K$ be a group homomorphism. Then ϕ is injective if and only if $\ker \phi = \{e\}$.

Proof. If ϕ is injective, then by the properties of homomorphisms $e \mapsto e$, and since ϕ is injective, there exist no other $g \in G$ such that $g \mapsto e$.

If $\ker \phi = \{e\}$, then we know idk ■

Chapter 7

The Class Equation and the Sylow Theorems

Note that this is chapter 23.

7.1 Class Equation

Definition 7.1: Conjugate

Let $a, b \in G$. We say a and b are conjugate in G if there is $x \in G$ such that $axa^{-1} = b$.

Definition 7.2: Conjugacy Class

The conjugacy class of a in G is the set $cl(a) = \{axa^{-1} \mid x \in G\}$. This is the set of all elements conjugate to a in G .

Note that in general, $cl(a) \neq G$. For example, $cl(e) = \{xe x^{-1} \mid x \in G\} = \{exx^{-1} \mid x \in G\} = \{e\}$. In fact, for $a \in Z(G)$, $cl(a) = \{a\}$ by the same logic.

Theorem 7.1

Conjugacy is an equivalence relation. Equivalently, the conjugacy classes in G partition G .

Proof. Let $a \in G$. Note that since $e \in G$, $ea e^{-1} = a \in cl(a)$. Therefore, $a \sim a$.

Let $a, b \in G$ such that $a \sim b$. It follows that there exists $x \in G$ such that $a = b x x^{-1}$. It follows that $x^{-1} a x = b$. Since $x^{-1} \in G$, $a \in cl(b)$, and $b \sim a$.

Let $a, b, c \in G$ such that $a \sim b$ and $b \sim c$. It follows that there exists $x \in G$ such that $a = b x x^{-1}$, and there exists $y \in G$ such that $b = c y y^{-1}$. Notice that $a = x c y y^{-1} x^{-1}$. Notice that $(xy)^{-1} = y^{-1} x^{-1}$. Since $xy \in G$, $c \in cl(a)$. Therefore, $a \sim c$, and $\sim$ is an equivalence relation. ■

Theorem 7.2

For $a \in G$, $|cl(a)| = [G : C(a)]$.

Proof. Since $C(a)$ is normal, note that

$$[G : C(a)] = |G/C(a)|.$$

Let $T : G/C(a) \rightarrow cl(a)$ be the map $xC(a) \mapsto xax^{-1}$. We wish to show T is a well-defined bijection.

First, we show T is well defined. Let $x, y \in G$ such that $xC(a) = yC(a)$. If $y = xg$ with $g \in C(a)$, then $yay^{-1} = xgag^{-1}x^{-1} = xagg^{-1}x^{-1} = xax^{-1}$.

Next, we show T is injective. Let $xax^{-1} = yay^{-1}$. It follows that $ax^{-1} = x^{-1}yay^{-1}$, and then $ax^{-1}y = x^{-1}ya$. Therefore, $x^{-1}y \in C(a)$, and $xC(a) = yC(a)$. Therefore, T is injective.

Finally, we show T is surjective. Let $xax^{-1} \in cl(a)$. There then exists $x \in G$, and thus $T(xC(a)) = xax^{-1}$. Therefore, T is surjective, and is thus a bijection. ■

Theorem 7.3 (The Class Equation)

Let G be a finite group, and let $a_1, \dots, a_r$ be representatives of the distinct conjugacy classes in G . Then

$$|G| = \sum_{i=1}^r |cl(a_i)| = \sum_{i=1}^r [G : C(a_i)].$$

This can be rewritten as

$$|G| = |Z(G)| + \sum_{a_i \notin Z(G)} [G : C(a_i)].$$

Proof. Since $a_1, \dots, a_r$ represent distinct conjugacy classes, we know $cl(a_i) \cap cl(a_j) = \emptyset$ for all $i \neq j$, and

$$\bigcup_{i=1}^r cl(a_i) = G.$$

Therefore, we can define a bijection

$$\phi : \bigcup_{i=1}^r cl(a_i) \rightarrow G$$

as the map

$$a \mapsto a.$$

It follows that

$$|G| = \left| \bigcup_{i=1}^r cl(a_i) \right| = \sum_{i=1}^r |cl(a_i)| = \sum_{i=1}^r [G : C(a_i)].$$

■

Lecture 10: Sylow Theorems

Tue 05 Nov 2024 09:31

Corollary 7.1

If G is a nontrivial finite group of order p^n for p prime, then $|Z(G)| > 1$.

Proof. See homework 8 problem 4a. ■

Corollary 7.2

Every group of order p^2 is abelian.

Proof. See homework 8 problem 4b. ■

7.2 Sylow Theorems

Recall Lagrange's theorem. If $H \leq G$ with G finite, then $|H|$ divides $|G|$. Remember that the converse is not true; that is, if a divides $|G|$, then there need not exist a subgroup of order a . The Sylow Theorems give us a partial converse to Lagrange's theorem.

Theorem 7.4 (Sylow's First Theorem, 1872)

Let G be a finite group and p a prime number. If p^k divides $|G|$, then G has a subgroup of order p^k .

Proof. Let G be a finite group. We will use proof by strong induction on $|G|$. Suppose that $|G| = 1$. Then the statement is true vacuously because there exists no such prime that divides $|G|$.

Now let the statement be true for all groups G such that $|G| < n$. We wish to prove it holds for $|G| = n$. Suppose p^k divides $|G|$. If G has a proper subgroup H with p^k dividing $|H|$, then we are done by induction.

Now suppose there exists no proper subgroup of this form. By the class equation,

$$|G| = |Z(G)| + \sum_{a_i \notin Z(G)} [G : C(a_i)],$$

where $a_1, \dots, a_r$ are the representatives of distinct conjugacy classes of G . For any i , it follows that

$$|G| = \frac{|G|}{|C(a_i)|} |C(a_i)|.$$

Since $C(a_i) \leq G$, by assumption, p^k does not divide $|C(a_i)|$. However, p^k divides $|G|$, and thus it divides $\frac{|G|}{|C(a_i)|} |C(a_i)|$. Thus, $p \mid [G : C(a_i)]$ since it is greater than 0 and the only divisors of p^k are powers of p . Therefore,

$$p \mid \left(\sum_{a_i \notin Z(G)} [G : C(a_i)] \right).$$

Therefore, p divides $|Z(G)|$. Since $Z(G)$ is abelian, by the Fundamental Theorem of Finite Abelian Groups, $Z(G)$ has an element of order p since one of the cyclic groups in its decomposition must have order divisible by p . Let this element be x , and consider $\langle x \rangle \leq G$. We know $|\langle x \rangle| = p$. We want to reduce the order of G using this group. We will use quotient groups. Note that $\langle x \rangle \triangleleft G$ since $x \in Z(G)$ and $\langle x \rangle \subseteq Z(G)$. The group $G/\langle x \rangle$ has order $|G|/p < G$. Furthermore, p^{k-1} divides $|G/\langle x \rangle|$. By our inductive hypothesis, there exists $H \leq G/\langle x \rangle$ such that $|H| = p^{k-1}$.

Exercise: Show $H = K/\langle x \rangle$ for some $K \leq G$. Construction is $K = \{g_i x^j\}$.

Then we have $|K| = |H|p$. Therefore, $|K| = p^k$. ■

Corollary 7.3 (Cauchy's Theorem)

Let G be a finite group and let p divide $|G|$, with p prime. Then G has an element of order p .

Proof. By Sylow's theorem, since p^1 divides $|G|$, G has a subgroup of order p . Since all groups of order p are cyclic, there exists an element of order p . ■

Definition 7.3: Sylow p -Subgroup

Let G be a finite group such that p^k divides $|G|$, for p prime. If $H \leq G$ is a subgroup such that $|H| = p^k$ is the largest power of p that divides $|G|$, then H is a Sylow p -subgroup of G .

Definition 7.4: Conjugate Subgroups

Let $H, K \leq G$. Then H and K are conjugate if there exists $g \in G$ such that $H = gKg^{-1}$.

Theorem 7.5 (Sylow's Second Theorem)

Let G be a finite group with $H \leq G$, and let $|H| = p^k$ for p a prime number. Then H is a subset (group) of some Sylow p -subgroup of G .

(Equivalent statement?) All Sylow p -subgroups are conjugate to each other.

Proof. Requires orbit-stabilizer theorem. ■

Theorem 7.6 (Sylow's Third Theorem)

Let G be a finite group with p a prime number such that $|G| = p^k m$ with $p \nmid m$. In other words, p^k is the largest power of p that divides $|G|$, so the Sylow p -subgroups have order p^k . Then the number of Sylow p -subgroups of G is equal to 1 (mod p) and divides m . Furthermore, all Sylow p -subgroups are conjugate.

Proof. Also requires orbit-stabilizer theorem. ■

Corollary 7.4

A Sylow p -subgroup is normal if and only if it's the only Sylow p -subgroup.

Proof. Let K be a Sylow p -subgroup of G . If $K \triangleleft G$, then $gKg^{-1} = K$ for all $g \in G$, so by Sylow's 3rd theorem, K is the only Sylow p -subgroup. If K is the only Sylow p -subgroup of G , then $gKg^{-1} = K$ for all $g \in G$ by Sylow's theorem, and $K \triangleleft G$. ■

Notation. Let n_p be the number of Sylow p -subgroups of G .

Theorem 7.7 (LOL WTF)

Every group of order 45 is abelian.

Proof. Let G be a group with $|G| = 45$. Notice $45 = 3^2 \cdot 5$. Thus, $3|45$. Let n_3 be the number of Sylow 3-subgroups of G . By Sylow's theorem, $n_3|5$ and $n_3 \equiv 1 \pmod{3}$. Therefore, $n_3 = 1$. Let n_5 be the number of Sylow 5-subgroups of G . Then $n_5|9$ and $n_5 \equiv 1 \pmod{5}$. Therefore, $n_5 = 1$.

Let P be the Sylow 3-subgroup, and let Q be the Sylow 5-subgroup of G . Then $P, Q \triangleleft G$. Therefore, $PQ = QP$, and PQ is a subgroup of G , and $P, Q \leq PQ$. By Lagrange's theorem, $|P|$ and $|Q|$ divide $|PQ|$. Since $|P| = 9$ and $|Q| = 5$, we know 9 and 5 divide $|PQ| \leq 45$. Therefore, $|PQ| = 45$, and $PQ = G$. Therefore, $G \cong P \times Q$.

Notice $|P| = 3^2$, so P is abelian by the class equation and the fundamental theorem of finite abelian groups. Notice $|Q| = 5$, so Q is cyclic and abelian. Therefore, $P \times Q$ is abelian, and G is abelian by isomorphism. ■

Theorem 7.8

Every group of order pq for p, q prime numbers, $p < q$, and $q \not\equiv 1 \pmod{p}$ is cyclic.

Proof. EFY ■

Theorem 7.9

The only group of order 255 is $\mathbb{Z}_{255}$.

Proof. use *Aut* and normalizers. ■

Remark

Keith Conrad expository notes are good to read.

Chapter 8

Rings

Lecture 11: Rings

Tue 12 Nov 2024 09:37

Definition 8.1: Ring

A set R is a ring if it has two closed binary operations $+$, $\cdot$ that satisfy the following axioms:

1. $a + b = b + a$ for all $a, b \in R$.
2. $a + (b + c) = (a + b) + c$ for all $a, b, c \in R$.
3. There exists an element $0 \in R$ such that $a + 0 = a$ for all $a \in R$.
4. For all $a \in R$, there exists $-a \in R$ such that $a + (-a) = 0$.
5. $(ab)c = a(bc)$ for all $a, b, c \in R$.
6. For all $a, b, c \in R$, $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$.

Definition 8.2: Group Ring

Let R be a commutative ring with unity, and let G be a finite group $G = \{g_1, \dots, g_n\}$. Then the group ring $R[G]$ is defined as

$$R[G] := \left\{ \sum_{i=1}^n r_i g_i \mid \forall i : r_i \in R \right\}.$$

Addition is componentwise, and multiplication is done as $(r_i g_i)(r_j g_j) = r_i r_j g_k$, where $g_i g_j = g_k$.

Definition 8.3: Commutative Ring

A commutative ring is a ring R such that $ab = ba$ for all $a, b \in R$.

Lecture 12: More Rings

Thu 14 Nov 2024 09:31

Remark

The p -adic fields have a strange topology where if two open sets intersect then they are equal.

Definition 8.4: Unity

A ring R has unity if there is $1 \in R$ such that $a1 = 1a = a$ for all $a \in R$.

Note. Some books only talk about rings with unity, and call a ring without unity a rng. A lot can be said about rings with unity that do not generalize. However, we are not doing so.

Definition 8.5: Integral Domain

A commutative ring R with unity is an integral domain if for every $a, b \in R$ such that $ab = 0$, then either $a = 0$ or $b = 0$. We say there are no zero divisors.

Definition 8.6: Domain

A commutative ring is a domain if there are no zero divisors.

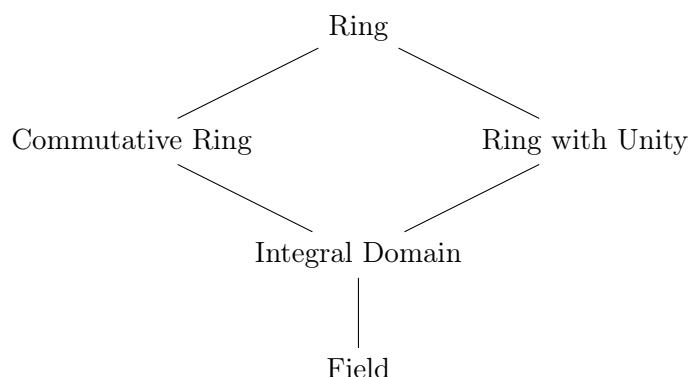
Definition 8.7: Field

A field is a commutative ring with unity, no zero divisors, and for all nonzero $a \in R$ there exists $a^{-1} \in R$ such that $aa^{-1} = 1$.

Definition 8.8: Division Ring

A division ring is a ring with unity and no nonzero divisors.

We are only concerned about the following categories.

**Definition 8.9: Unit**

Let R be a ring with $a \in R \setminus \{0\}$. If a has a multiplicative inverse, we say a is a unit of R . We denote the set of units in R as $U(R)$.

Definition 8.10: Zero Divisor

A nonzero element $a \in R$ in a commutative ring is called a zero divisor if there is a nonzero element $b \in R$ such that $ab = 0$.

Remark

Proof by googling it: a noncommutative ring with unity can also have zero divisors. A rng has left/right zero divisors.

Recall that $M_2(\mathbb{C})$ is a ring. It only has unity though. We want to find a matrix ring that has some properties. Let

$$1 = I_2, \quad \hat{\mathbf{i}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \hat{\mathbf{j}} = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \quad \hat{\mathbf{k}} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}.$$

Define

$$\mathbb{H} = \left\{ a1 + b\hat{\mathbf{i}} + c\hat{\mathbf{j}} + d\hat{\mathbf{k}} \mid a, b, c, d \in \mathbb{R} \right\}.$$

We call $\mathbb{H}$ the set of **quaternions**. We will show $\mathbb{H}$ is a non-commutative ring with unity, but without zero divisors. Recall that

$$\begin{aligned} \hat{\mathbf{i}}^2 = \hat{\mathbf{j}}^2 = \hat{\mathbf{k}}^2 = \hat{\mathbf{i}}\hat{\mathbf{j}}\hat{\mathbf{k}} &= -1, \\ \hat{\mathbf{i}}\hat{\mathbf{j}} &= \hat{\mathbf{k}}, \quad \hat{\mathbf{j}}\hat{\mathbf{k}} = \hat{\mathbf{i}}, \quad \hat{\mathbf{k}}\hat{\mathbf{i}} = \hat{\mathbf{j}}, \\ \hat{\mathbf{j}}\hat{\mathbf{i}} &= -\hat{\mathbf{k}}, \quad \hat{\mathbf{k}}\hat{\mathbf{j}} = -\hat{\mathbf{i}}, \quad \hat{\mathbf{i}}\hat{\mathbf{k}} = -\hat{\mathbf{j}}. \end{aligned}$$

We say $\mathbb{H}$ is anti-commutative. Note that if we multiply a quaternion $\mathbf{h}$ with its conjugate $\bar{\mathbf{h}}$, we have

$$(a1 + b\hat{\mathbf{i}} + c\hat{\mathbf{j}} + d\hat{\mathbf{k}})(a1 - b\hat{\mathbf{i}} - c\hat{\mathbf{j}} - d\hat{\mathbf{k}}) = a^2 + b^2 + c^2 + d^2.$$

Therefore, these two things multiply to zero if and only if $a, b, c, d = 0$. We have now shown for $\mathbf{h} \neq 0$, it and its conjugate can be divided by this value. If $a1 + b\hat{\mathbf{i}} + c\hat{\mathbf{j}} + d\hat{\mathbf{k}} \neq 0$, then

$$(a1 + b\hat{\mathbf{i}} + c\hat{\mathbf{j}} + d\hat{\mathbf{k}}) \left(\frac{a1 - b\hat{\mathbf{i}} - c\hat{\mathbf{j}} - d\hat{\mathbf{k}}}{a^2 + b^2 + c^2 + d^2} \right) = 1.$$

This tells us that every element that is nonzero is a unit. It remains to be shown **that units are not zero divisors**.

Theorem 8.1 (Properties of Rings)

Let R be a ring, and let $a, b \in R$. Then

- $a0 = 0a = 0$.
- $a(-b) = (-a)b = -(ab)$.
- $(-a)(-b) = ab$.
- $a(b - c) = ab - ac$.

If R has unity, then

- $(-1)a = -a$.
- $(-1)(-1) = 1$.
- The unity 1 is unique.

Proof. (1)

$$a0 = a(0 + 0) = a0 + a0.$$

Since there are inverses under addition, $a0 - a0 = a0 + a0 - a0$ implies $a0 = 0$. Same for $0a$.

(2)

$$a(-b) + ab = a(-b + b) = a0 = 0.$$

Since inverses are unique under addition, $a(-b) = -ab$. Similarly,

$$(-a)b + ab = (-a + a)b = 0b = 0.$$

Therefore, $a(-b) = (-a)b = -ab$.

(3)

$$(-a)(-b) + ((-a)b) = (-a)(-b + b) = (-a)0 = 0.$$

Since inverses are unique under addition, $(-a)(-b) = -((-a)b) = -(-ab) = ab$.

(4)

$$a(b - c) = a(b + (-c)) = ab + a(-c).$$

By property (2), $a(-c) = -ac$. Therefore, $a(b - c) = ab - ac$.

(5)

$$a + (-1)a = 1a - 1a = (1 - 1)a = 0a = 0.$$

Since inverses are unique under addition, $(-1)a = -a$.

(6)

$$(-1)(-1) - 1 = (-1)(-1) + (-1)1 = (-1)(-1 + 1) = (-1)0 = 0.$$

Since inverses are unique under addition, $(-1)(-1) = -(-1) = 1$.

(7) Let e_1, e_2 be identities in R . Since e_1 is an identity, $e_2 = e_1e_2$. Since e_2 is an identity, $e_1 = e_1e_2$.

Therefore $e_1 = e_2$. ■

Proposition 8.1

Let R be a ring, and let $a \in R$ be a unit. Then a is not a zero divisor.

Proof. Let $a \in R$ be a unit, and let $ab = 0$. Then $a^{-1}ab = a^{-1}0$. By properties of rings, $a^{-1}0 = 0$. Furthermore, $a^{-1}ab = 1b = b$. Therefore, $b = 0$, and a is not a zero divisor. ■

Definition 8.11: Subring

Let $(R, +, \cdot)$ be a ring with subset $S \subseteq R$. Then S is called a subring if $(S, +, \cdot)$ is a ring under the inherited operations.

Proposition 8.2

Let R be ring with $S \subseteq R$. Then S is a subring if and only if

1. $S \neq \emptyset$,
2. $rs \in S$ for all $r, s \in S$,
3. $r - s \in S$ for all $r, s \in S$.

Proof. h ■

For example, consider the ring $M_2(\mathbb{R})$, and consider the subset $S \subseteq M_2(\mathbb{R})$ defined as

$$S := \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \mid a, b, c \in \mathbb{R} \right\}.$$

We will show S is a subring of $M_2(\mathbb{R})$. Note that $I \in S$, so $S \neq \emptyset$. Furthermore, note that

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \begin{pmatrix} x & y \\ 0 & z \end{pmatrix} = \begin{pmatrix} ax & ay + bz \\ 0 & cz \end{pmatrix} \in S.$$

Finally, note that

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} - \begin{pmatrix} x & y \\ 0 & z \end{pmatrix} = \begin{pmatrix} a - x & b - y \\ 0 & c - z \end{pmatrix}.$$

Therefore, S is a subring of $M_2(\mathbb{R})$.

— **Remark** —

Let R be a ring with unity. Note that $\{0\} \subseteq R$ is a ring without unity. Therefore, subrings do not need to have the unity.

Another good example of a ring is the Gaussian Integers.

— **Definition 8.12: Gaussian Integers** —

The gaussian integers are the set $\mathbb{Z}[i]$ defined as

$$\mathbb{Z}[i] := \{a + bi \mid a, b \in \mathbb{Z}\} \subset \mathbb{C}.$$

We claim $\mathbb{Z}[i]$ is an integral domain, but not a field. Let $\alpha = a + bi \in \mathbb{Z}[i]$, and suppose α is a unit. Then there exists $\beta = c + di \in \mathbb{Z}[i]$ such that $\alpha\beta = 1$. We want to show this does not happen in general. Consider the conjugate $\overline{\alpha}\overline{\beta} = 1$. We then have $\alpha\beta\overline{\alpha}\overline{\beta} = 1$. We have

$$(a + bi)(a - bi)(c + di)(c - di) = (a^2 + b^2)(c^2 + d^2) = 1.$$

We now have the product of two positive integers are 1. This happens if and only if both integers are 1. That is, $a^2 + b^2 = c^2 + d^2 = 1$. Then $a = \pm 1$ and $b = 0$, or vice versa. These are the only cases where α has a unity β . Therefore, $\mathbb{Z}[i]$ is **not** a field.

Chapter 9

Integral Domains

Lecture 13: Integral Domains

Tue 19 Nov 2024 09:32

Recall the definition.

Definition 9.1: Integral Domain

A ring R is an integral domain if it has unity, is commutative, and has no zero divisors.

Proposition 9.1 (Cancellation Law)

Let D be a commutative ring with unity. Then D is an integral domain if and only if for all nonzero $a \in D$ with $ab = ac$, we have $b = c$.

Proof. Let D be an integral domain, and let $ab = ac$ for $a, b, c \in D$ and a nonzero. We have

$$ab - ac = ac - ac = 0.$$

By the distributive property,

$$a(b - c) = 0.$$

Since a is nonzero and D has no zero divisors, it follows that $b - c = 0$, and thus $b = c$.

Now let D be a commutative ring with unity, and let $ab = ac$ imply that $b = c$ for all $a, b, c \in D$ and a nonzero. Suppose $ad = 0$ for some $b \in D$. Since $a0 = 0$, we have $ad = a0$. Therefore, since a nonzero, we have $d = 0$. ■

Theorem 9.1

Every finite integral domain is a field.

Proof. Let D be a finite integral domain, and let $D^* = D \setminus \{0\}$. Define $\lambda_a : D^* \rightarrow D^*$ as the map $d \mapsto ad$. We want to show that $1 \in \lambda_a(D^*)$. We note that λ_a is well defined because $a, b \neq 0$ implies $ab \neq 0$ because there are no zero divisors. We will now show λ_a is injective. Let $ab = ac$. It follows by the cancellation law that $b = c$, and thus λ_a is injective. Since D^* is finite, the map is automatically a bijection, and $1 \in D^*$ implies $1 \in \lambda_a(D^*)$. Therefore, a has an inverse element. ■

Corollary 9.1

$\mathbb{Z}_p$ is a field, for p prime.

There are fields of nonprime order. For example, $\mathbb{Z}_3[i]$ has order 9.

Definition 9.2: Characteristic

The characteristic of a ring R is the smallest $n \in \mathbb{N}$ such that $nr = 0$ for all $r \in R$. If no such integer exists, we say R has characteristic 0. We denote the characteristic as $\text{char}(R)$.

Remark

When we have a ring R with $r \in R$ and $a \in \mathbb{Z}$, we write

$$ar = \sum_{i=1}^a r.$$

However, since R may not be $\mathbb{Z}$, we cannot applying the ring axioms to a or other elements of $\mathbb{Z}$.

Theorem 9.2

Let R be a ring with unity. Then $\text{char}(R) \neq 0$ implies $\text{char}(R) = |1|_+$, for $|\cdot|_+$ the additive order. Moreover, $|1|_+ = \infty$ implies $\text{char}(R) = 0$.

Proof. Let R be a ring with unity. Suppose $\text{char}(R) = m < \infty$. Then $|1| \leq m$. Suppose $|1|_+ = n$. Then $n1 = 0$ by definition. Thus for all $r \in R$, we have

$$nr = n(1r) = (n1)r = 0r = 0.$$

Therefore, $\text{char}(R) = n$.

Now suppose $|1|_+ = \infty$. Then there exists no n such that $n1 = 0$, and thus the characteristic is 0 by definition. ■

Theorem 9.3

The characteristic of an integral domain is either prime or 0.

Proof. Let D be an integral domain. Suppose $\text{char}(D) = n \neq 0$. Since D has unity, $n = |1|_+$. If n is not prime, then $n = ab$ for some $a, b \in \mathbb{N}$ with $1 < a, b < n$. We then have $|1|_+ = ab$, and $ab1 = 0$. Therefore, $(a1)(b1) = 0$. Since D is an integral domain, it follows that either $a1$ or $b1$ is 0. Therefore, $|1|_+$ either equals a or b , and thus does not equal n . This is a contradiction, implying $|1|_+$ is prime. ■

Chapter 10

Ideals and Quotient Rings

Lecture 14: Ideals and Quotient Rings

Thu 21 Nov 2024 09:31

An ideal of a ring is analagous to a normal subgroup of a group.

Definition 10.1: Ideal

An ideal I in a ring R is a subring $I \subseteq R$ such that if $a \in I$ and $r \in R$, $ar, ra \in I$. That is, $rI \subseteq I$ and $Ir \subseteq I$ for all $r \in R$.

Every ring R has at least two ideals, R and $\{0\}$. These are the trivial ideals. If R is a ring with unity, and $I \triangleleft R$ is an ideal with the unity of R contained in I , then $I = R$ because $r1 \in I$ for all r by definition of an ideal.

Definition 10.2

Let R be a commutative ring with unity, and let $\langle a \rangle = \{ar \mid r \in R\}$.

Theorem 10.1

Let $a \in R$ be a commutative ring with unity. Then $\langle a \rangle$ is an ideal in R .

Proof. Note that $a1 = a \in \langle a \rangle$. Note that $ar_1 - ar_2 = a(r_1 - r_2) \in \langle a \rangle$. Finally, note $r_2ar_1 = ar_1r_2 \in \langle a \rangle$. Finally, $ar_1ar_2 = a(r_1ar_2) \in \langle a \rangle$. This is an ideal ■

Definition 10.3: Principal

An ideal I is **principal** if $I = \langle a \rangle$ for some $a \in R$.

For example, every ideal in $\mathbb{Z}$ is principal. If I is nontrivial, I must contain some $m \in \mathbb{Z}^+$. Then there must be a least positive integer $n > 0 \in I$. Let $a \in I$. We have $a = nq + r$ for some $q, r \in \mathbb{Z}$ and $0 \leq r < n$. Then $r = a - nq$. Since $a, nq \in I$, $r \in I$. Therefore, $r = 0$, and $a = nq \in \langle n \rangle$. Therefore, $\langle n \rangle = I$.

Note that not all ideals are principal. Consider $I = \{2a + xb \mid a, b \in \mathbb{Z}\}$, an ideal of $\mathbb{Z}[x]$. This ideal is generated by two elements.

Theorem 10.2

Let I be an ideal of R . The quotient group R/I is a ring with multiplication defined by $(r + I)(s + I) = rs + I$.

Proof. R/I is a quotient group because $(R, +)$ is abelian. We need to prove it now forms a ring. Let $r + I, s + I, t + I \in R/I$. First, we show multiplication is well defined. Let $r + I = s + I$. Since $r' \in r + I$ by properties of cosets, there exists $a \in I$ such that $r' = r + a$. Similarly, there exists $b \in I$ such that $s' = s + b$. We now have

$$(r' + I)(s' + I) = r's' + I = (r + a)(s + b) + I = (rs + as + rb + ab) + I.$$

We want to show $rs + as + rb + ab \in I$. Recall that $a, b \in I$, so any product with them is in I . Therefore, $rs, as, rb, ab \in I$, and since I is a ring, $rs + as + rb + ab \in I$. Therefore, $r's' + I = rs + I$.

Closure is trivial. Consider associativity.

$$(r + I)((s + I)(t + I)) = (r + I)(st + I) = r(st) + I = (rs)t + I = ((r + I)(s + I))(t + I).$$

Therefore, R/I is associative. Now consider distributivity.

$$(r + I)(s + I + t + I) = (r + I)((s + t) + I) = r(s + t) + I = (rs + rt) + I = (rs + I) + (rt + I).$$

The proof is the same on the other side. Therefore, R/I is a ring. ■

Definition 10.4: Quotient Ring

Let R be a ring and let $I \subseteq R$ be an ideal. Then R/I is a ring with addition defined as

$$(r + I) + (s + I) = (r + s) + I,$$

and multiplication defined as

$$(r + I)(s + I) = rs + I.$$

For example, somehow $\mathbb{Z}_3[x]/\langle x^2 + 1 \rangle$ is a quotient ring. The identity is $\langle x^2 + 1 \rangle$. The multiplicative identity is $1 + \langle x^2 + 1 \rangle$, and the other elements are of the form $p(x) + \langle x^2 + 1 \rangle$. Also, consider

$$x^2 + \langle x^2 + 1 \rangle = 2 + \langle x^2 + 1 \rangle.$$

Since $x^2 + 1 \in \langle x^2 + 1 \rangle$ is the identity, $x^2 - (x^2 + 1) + \langle x^2 + 1 \rangle = -1 + \langle x^2 + 1 \rangle$. Then $-1 \equiv 2 \pmod{3}$.

Lecture 15: Ideals and Homomorphisms

Tue 26 Nov 2024 09:30

Definition 10.5: Maximal Ideal

An ideal M in a ring R is maximal if $M \neq R$ and the only ideal strictly containing M is R .

Rings can have multiple maximal ideals. For example, $p\mathbb{Z}$ is a maximal ideal of $\mathbb{Z}$ for all prime p . Moreover, Krull's theorem states maximal ideals always exist.

Theorem 10.3

Let R be a commutative ring with unity, and let $M \subset R$ be an ideal. Then M is maximal if and only if R/M is a field.

Proof. done in homework 10 ■

Note that $\mathbb{Z}/p\mathbb{Z}$ is a field, proving our above statement that $\mathbb{Z}$ has multiple maximal ideals.

Definition 10.6: Prime Ideal

A proper ideal P of a commutative ring R is **prime** if $ab \in P$ if and only if $a \in P$ or $b \in P$.

An important property of prime numbers is that for a prime p , if $p|ab$, then $p|a$ or $p|b$. This is the same idea, but now with ideals instead of like numbers.

Proposition 10.1

Let R be a commutative ring with unity. Then P is a prime ideal of R if and only if R/P is an integral domain.

Proof. also done in homework 10 ■

Corollary 10.1

Any maximal ideal is prime, but prime ideals are not necessarily maximal.

Proof. Any field is an integral domain, so it follows immediately from the previous theorem and proposition. ■

Corollary 10.2

A commutative ring with unity ring R is a field if and only if $\langle 0 \rangle$ is a maximal ideal.

Proof. done in the homework, except this is not a corollary in a homework but a lemma to the previous theorems. ■

Proposition 10.2

A commutative ring with unity R is an integral domain if and only if $\langle 0 \rangle$ is a prime ideal.

Proof. Let $ab = 0$. Then either a or b is 0. Therefore, R is an integral domain. ■

Chapter 11

Ring Homomorphisms

Definition 11.1: Ring Homomorphism

Let R, S be rings, and let $\phi : R \rightarrow S$. Then ϕ is a ring homomorphism if and only if for all $a, b \in S$,

$$\phi(ab) = \phi(a)\phi(b) \quad \text{and} \quad \phi(a + b) = \phi(a) + \phi(b).$$

Definition 11.2: Ring Isomorphism

Let R, S be rings and let $\phi : R \rightarrow S$ be a ring homomorphism. Then ϕ is an isomorphism if and only if it is bijective. We write $R \cong S$ to denote that R is isomorphic to S .

Definition 11.3: Kernel

Let R, S be rings and let $\phi : R \rightarrow S$ be a ring homomorphism. Define the kernel of ϕ as the set

$$\ker \phi = \{r \in R \mid \phi(r) = 0\}.$$

Proposition 11.1

Let R, S be rings and let $\phi : R \rightarrow S$ be a ring homomorphism. Then $\ker \phi$ is an ideal of R .

Proof. Done in homework. I will take that $\ker \phi$ is a ring to be trivial. Let $r \in \ker \phi$. Note that

$$\phi(ar) = \phi(a)0 = 0.$$

Therefore, $ar \in \ker \phi$. ■

Proposition 11.2

Let R be a ring with I an ideal. Define $\phi : R \rightarrow R/I$ as the map $r \mapsto r + I$. Then $\ker \phi = I$.

Proof. Let $r \in \ker \phi$. It follows that $\phi(r) = r + I = I$. By properties of cosets, $r \in I$. Now let $r \in I$. It follows by properties of cosets that $\phi(r) = r + I = I$. Therefore, $r \in \ker \phi$. ■

Remark

We call $\phi : R \rightarrow R/I$ defined as $r \mapsto r + I$ the **canonical homomorphism**.

Here's an interesting ring homomorphism. Let ϕ_α be a linear functional. That is, let $\phi_\alpha : C[a, b] \rightarrow \mathbb{R}$ be the map $f \mapsto f(\alpha)$. Then ϕ_α is a ring homomorphism for all $\alpha \in \mathbb{R}$.

Theorem 11.1 (Properties)

Let $\phi : R \rightarrow S$ be a ring homomorphism.

- If R is a commutative ring, $\phi(R)$ is a commutative ring.
- $\phi(0) = 0$.
- Let R be a ring with unity. If ϕ is surjective, then $\phi(1) = 1$.
- If R is a field and $\phi(R) \neq \{0\}$, then $\phi(R)$ is a field.

Proof. trivial. ■

Theorem 11.2 (First Isomorphism Theorem)

Let R, S be rings and let $\psi : R \rightarrow S$ be a ring homomorphism. Then the map $\eta : R/\ker \psi \rightarrow \psi(R)$ defined as $r + \ker \psi \mapsto \psi(r)$ is a unique isomorphism. Moreover, if $\phi : R \rightarrow R/\ker \psi$ is the canonical isomorphism theorem, then $\psi = \eta \circ \phi$.

Theorem 11.3 (Second Isomorphism Theorem)

Let R be a ring and let I, J be ideals of R . Then

$$I/I \cap J \cong (I + J)/J.$$

Theorem 11.4 (Third Isomorphism Theorem)

Let R be a ring and let I, J be ideals of R with $J \subset I \subset R$. Then

$$R/I \cong \frac{R/J}{I/J}.$$

Theorem 11.5 (Correspondence Theorem)

Problem 5a of hw 10? idk

Lecture 16: Group Actions

Tue 03 Dec 2024 09:30

Remark

In a PID, an ideal is prime iff it cannot be factored. This generalizes nicely to polynomial rings. Consider a polynomial $x^2 + 1 \in \mathbb{R}[x]$ that cannot be factored in $\mathbb{R}$. Note that $\mathbb{R}[x]/\langle x^2 + 1 \rangle \cong \mathbb{R}[i] = \mathbb{C}$. This is the splitting field of the polynomial, so if you have an irreducible polynomial the ring mod the ideal may be the splitting field. If its reducible u have to reduce it first i think

Chapter 12

Group Actions

Since these aren't in the book, we will use the notes by Keith Conrad in his expository papers. First, we will motivate the definitions. On the first day, we considered symmetries of regular polygons, such as the square. These were given by the dihedral groups. For example, D_4 has 8 elements, and $D_4 \leq S_4$. Then the symmetries of the square are permutations of the set of four vertices. We want to generalize this idea of groups given as permutations of sets.

Definition 12.1: Group Action

An **action** of a group G on a set X is the choice for each $g \in G$ of a permutation $\Pi_g : X \rightarrow X$ such that

- Π_e is the identity, meaning $\Pi_e(x) = x$ for all $x \in X$.
- For every $g_1, g_2 \in G$, $\Pi_{g_1} \circ \Pi_{g_2} = \Pi_{g_1 g_2}$.

In our example, the group acting is D_4 , and the set is the set of 4 vertices.

Lecture 17: Group Actions

Thu 05 Dec 2024 09:30

For example, consider D_n acting on $\{1, \dots, n\}$ by arranging $\{1, \dots, n\}$ as the vertices of a regular n -gon. This is a group action. We can also consider a group G acting on itself by left multiplication; $\Pi_g : G \rightarrow G$ defined as $a \mapsto ga$. Notice this is basically just Cayley's theorem. One more example is that the set S_n acts on the set $X := \{1, \dots, n\}$ by $\Pi_\sigma(i) = \sigma(i)$ for all $i \in X$.

Proposition 12.1

Let G be a group that acts on a set X . Then the group of permutations on the set X given by the group action is a group.

Proof. Multiplication follows equivalently to the group G because of the second property of group actions, and the identity permutation is guaranteed to be defined properly for inverses to exist, so this proof is very trivial. ■

Notation. When our reference to Π_g is clear, we write $\Pi_g(x)$ as $g \cdot x$, or in some places, gx .

Notation. If G acts on X , we write $G \curvearrowright X$ or $G \curvearrowleft X$.

Theorem 12.1

Let $G \curvearrowright X$. If $x \in X$, $g \in G$, and $y = g \cdot x$, then $x = g^{-1} \cdot y$. If $x \neq x'$, then $g \cdot x \neq g \cdot x'$.

Proof. $g^{-1} \cdot y = g^{-1}(g \cdot x) = g^{-1}g \cdot x = e \cdot x = x$. Also, if $g \cdot x = g \cdot x'$, then $g^{-1}g \cdot x = g^{-1}g \cdot x'$, and thus $e \cdot x = e \cdot x'$, a contradiction. ■

Notation. $\text{Sym}(X)$ is defined as the set of permutations of the set X . It is just a representation of $S_{|X|}$ given the precise elements of X .

Theorem 12.2

Actions of a group G on a set X are the same as group homomorphisms from G to $\text{Sym}(X)$.

$$G \curvearrowright X \rightsquigarrow \phi : G \rightarrow \text{Sym}(X).$$

Proof. no proof! ■

For example, consider $\text{GL}_2(\mathbb{R}) \curvearrowright \mathbb{R}^2$. The action is basically for any $A \in \text{GL}_2(\mathbb{R})$ and $\mathbf{x} \in \mathbb{R}^2$, we have $\Pi_A(\mathbf{x}) = A\mathbf{x}$. Since $\text{GL}_2(\mathbb{R}) = \text{Iso}(\mathbb{R}^2)$, this holds. These sets are infinite and it still behaves very nicely. As another example, $S_n \curvearrowright R[x_1, \dots, x_n]$ for R a ring. For $\sigma \in S_n$, the action is

$$\sigma \cdot f(x_1, \dots, x_n) = f(x_{\sigma(1)}, \dots, x_{\sigma(n)}).$$

The map σ just acts on the indices of the variables. A specific example of this example is the following: consider S_3 and $\mathbb{Z}[x_1, x_2, x_3]$. Let $(12) \in S_3$. We have

$$(12) \cdot (x_1^2 + x_2 + x_3) = x_2^2 + x_1 + x_3.$$

The polynomials fixed by this action are called **symmetric polynomials**.

Definition 12.2: Symmetric Polynomials

Let $S_n \curvearrowright R[x_1, \dots, x_n]$. The set of all $p(x_1, \dots, x_n)$ such that for all $\sigma \in S_n$, the polynomial is fixed $\sigma \cdot p(x_1, \dots, x_n) = p(x_1, \dots, x_n)$, is called the set of **symmetric polynomials**.

For example, the set of symmetric polynomials in $\mathbb{Z}[x_1, x_2, x_3]$ look like the set generated by

$$\{x_1x_2x_3, x_1x_2 + x_1x_3 + x_2x_3, x_1 + x_2 + x_3\}.$$

Definition 12.3

Let $G \curvearrowright X$. For each $x \in X$, the orbit of x is

$$\text{Orb}_x = \{g \cdot x \mid g \in G\} \subseteq X.$$

The stabilizer of x is the set

$$\text{Stab}_x = \{g \in G \mid g \cdot x = x\} \subseteq G.$$

We call x a **fixed point** if $\text{Stab}_x = G$.

Note that the set of symmetric polynomials in $R[x_1, \dots, x_n]$ is just the set of all fixed points in $R[x_1, \dots, x_n]$.

Consider $\text{GL}_2(\mathbb{R}) \curvearrowright \mathbb{R}^2$. The orbit

$$\text{Orb}_{(0,0)} = \left\{ M \cdot \begin{pmatrix} 0 \\ 0 \end{pmatrix} \mid M \in \text{GL}_2(\mathbb{R}) \right\} = \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}.$$

The stabilizer

$$\text{Stab}_{(0,0)} = \left\{ M \in \text{GL}_2(\mathbb{R}) \mid M \cdot \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\} = \text{GL}_2(\mathbb{R}).$$

For $G \curvearrowright G$, for all $x \in G$, $\text{Orb}_x = G$, and $\text{Stab}_x = \{e\}$.

Consider now $G \curvearrowright G$ by conjugation; that is, $g \cdot a = gag^{-1}$. The orbit of any element a is the set

$$\text{Orb}_a = \{gag^{-1} \mid g \in G\} = cl(a).$$

The stabilizer of a is the set

$$\text{Stab}_a = \{g \in G \mid gag^{-1} = a\} = C(a).$$

Finally, a is a fixed point if and only if $C(a) = G$, and thus $a \in Z(G)$.

Theorem 12.3 (Fundamental Theorem of Group Actions)

Let G be a group, X be a set, and let $G \curvearrowright X$. Then the following are true.

1. Different orbits are disjoint, so orbits partition X .
2. For all $x \in X$, the stabilizer Stab_x is a subgroup of G . Also, $\text{Stab}_{gx} = g \text{Stab}_x g^{-1}$.
3. For all $x \in X$, there exists a bijection

$$G / \text{Stab}_x \rightarrow \text{Orb}_x.$$

In particular, $|G| / |\text{Stab}_x| = |\text{Orb}_x|$. Equivalently, $|G| = |\text{Orb}_x| |\text{Stab}_x|$, the orbit stabilizer theorem.